Sunyaev–Zeldovich effect: $T_{photon} \ll T_{electron}$
(T_γ) (T_e)

Suppose $n(\tilde{x})$ is close to Planck distribution

$$n(\omega) = \frac{1}{e^{\frac{\hbar\omega}{k_B T_\gamma}} - 1} \Rightarrow n(\tilde{x}) = \frac{1}{e^{\tilde{x}} - 1} \quad \text{where } \tilde{x} = \frac{\hbar\omega}{k_B T_\gamma}$$

T_γ = photon temperature.

Then $n + n^2 = n(1+n) = \dfrac{e^{\tilde{x}}}{(e^{\tilde{x}}-1)^2}$.

On the other hand $\dfrac{\partial n}{\partial \tilde{x}} = -\dfrac{e^{\tilde{x}}}{(e^{\tilde{x}}-1)^2} = -n(1+n)$

Note however that in Kompaneets equation

$$\frac{\partial n}{\partial t_e} = \frac{k_B T_e}{m_e c^2} \frac{1}{x^2} \frac{\partial}{\partial x}\left[x^4 \left(\frac{\partial n}{\partial x} + n + n^2\right)\right]$$

$x = \dfrac{\hbar\omega}{k_B T_e} = \dfrac{\hbar\omega}{k_B T_\gamma} \dfrac{T_\gamma}{T_e} = \dfrac{T_\gamma}{T_e}\tilde{x}$. Here T_e = electron temperature

Therefore $\dfrac{\partial n}{\partial x} = \dfrac{\partial n}{\partial \tilde{x}} \dfrac{\partial \tilde{x}}{\partial x} = \dfrac{T_e}{T_\gamma} \dfrac{\partial n}{\partial \tilde{x}} = -\dfrac{T_e}{T_\gamma} n(1+n)$.

That is, $n(1+n) = n + n^2 = -\dfrac{T_\gamma}{T_e} \dfrac{\partial n}{\partial x}$.

Thus for the situation when $T_\gamma \ll T_e$ (which is the case for cosmic microwave background and "hot" electrons in galaxy clusters)

$$n + n^2 \ll -\frac{\partial n}{\partial x}$$

$\Rightarrow$ Kompaneets equation can be approximated as

$$\frac{\partial n}{\partial t_e} = \frac{k_B T_e}{m_e c^2} \frac{1}{x^2} \frac{\partial}{\partial x}\left[x^4 \left(\frac{\partial n}{\partial x} - \cancel{\frac{T_\gamma}{T_e}\frac{\partial n}{\partial x}}\right)\right]$$

neglected as $\ll 1$.

Take $y = \dfrac{k_B T_e}{mc^2} t_e \Rightarrow \dfrac{\partial n}{\partial t_e} = \dfrac{\partial n}{\partial y}\dfrac{\partial y}{\partial t_e}$ and $\dfrac{\partial y}{\partial t_e} = \dfrac{k_B T_e}{m e^2}$.

Thus obtain $\frac{\partial n}{\partial y} = \frac{1}{x^2} \frac{\partial}{\partial x}\left(x^4 \frac{\partial n}{\partial x}\right)$

Using $n(x) \cong \frac{1}{e^x - 1}$ can simplify RHS as $-x \frac{e^x}{(e^x-1)^2}\left(4 - x\frac{e^x+1}{e^x-1}\right)$

i.e. $$\frac{\partial n}{\partial y} = -x e^x n^2 \left(4 - x \coth\frac{x}{2}\right).$$

Then for small y we can obtain

$$\frac{\Delta n}{n} = -y\, x\, e^x n \left(4 - x \coth\frac{x}{2}\right).$$

This is subject of HW05, Prob. 3.

* As a result, photons are described by modified Planck distribution which is shifted to <u>higher</u> frequencies.

Non-linear regime (Zeldovich–Levich 1969)

Consider now the situation with $T_{photon} = T_\gamma \gg T_{electron}$

Assume that in the regime of low frequencies

photon occupation numbers are <u>large</u>, $n(\omega) \gg 1$.

That is $n(\omega) = \dfrac{1}{e^{\frac{\hbar\omega}{kT_\gamma}} - 1} \approx \dfrac{kT_\gamma}{\hbar\omega} + \ldots \gg 1.$

Than $n^2 \gg n$. We also assume $n^2 \gg \dfrac{\partial n}{\partial x}$.

$$\Rightarrow \quad \frac{\partial n}{\partial t} = \frac{1}{x^2}\frac{\partial}{\partial x}\left[x^4\left(n^2 + \cancel{n} + \cancel{\frac{\partial n}{\partial x}}\right)\right]$$

Let $f(x,t) = x^2 n(x,t) \Rightarrow \dfrac{\partial f}{\partial t} = \dfrac{\partial f^2}{\partial x} = 2f\dfrac{\partial f}{\partial x}$ non-linear diff. eq.

$$\cancel{\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2} \Rightarrow \text{characteristics } \frac{df}{dt} = 0 \text{ along } \frac{dx}{dt} = f} \quad ?$$

$$\cancel{x = F(f) - 2tf}$$

$$\cancel{\frac{dx}{dt} = F'\left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial t}\right) - 2f - 2t\frac{\partial f}{\partial t} =}$$

$$\cancel{= (F' - 2t)\frac{\partial f}{\partial t} +}$$

$$f = x^2 h$$

$$\Rightarrow x + 2ft = \phi[f] \quad \text{where } \phi = F^{-1}.$$

$$f(x,t) = F(x + 2ft), \quad f(x,0) = F(x).$$

$$\frac{\partial f}{\partial t} = F'\left(2f + 2t\frac{\partial f}{\partial t}\right) = F' \cdot 2f + 2tF'\frac{\partial f}{\partial t}$$

$$\Rightarrow \frac{\partial f}{\partial t} = \frac{2fF'}{1 - 2tF'}$$

This page is really sketchy. Sorry.

$$\frac{\partial f}{\partial x} = F'\left(1 + 2t\frac{\partial f}{\partial x}\right) = F' + 2tF'\frac{\partial f}{\partial x}$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{F'}{1 - 2tF'}$$

$$\cancel{\text{let } n(k,0) = \frac{1}{e^{k/\theta} - 1} \Rightarrow n(x,0) = \frac{1}{e^{x/x_0} - 1}}$$

$$\cancel{f(x,0) = \frac{x^2}{e^{\frac{x}{x_0}} - 1} \Rightarrow F(x) = \frac{x^2}{e^{\frac{x}{x_0}} - 1}}$$

$$\cancel{\Rightarrow F(x + 2ft) = f(x,t) = \frac{(x + 2ft)^2}{e^{\frac{x+2ft}{x_0}} - 1}}$$

$$\cancel{x + 2ft = x_0}$$

$$\cancel{x = x_0 - 2ft}$$

$$\cancel{4f^2\tau^2 = e^{\frac{2f\tau}{x_0}} - 1 \approx \frac{2f\tau}{x_0} \Rightarrow \tau = \frac{1}{2fx_0}}$$

$$\cancel{\Rightarrow \text{at } x = 0 \quad f(0,\tau) = F(2f\tau) = \frac{(2f\tau)^2}{e^{\frac{2f\tau}{x_0}} - 1}}$$

let $n(x,0) = \dfrac{1}{e^{\alpha x}-1} = \dfrac{1}{\alpha x + \frac{1}{2}\alpha^2 x^2}$ for $\alpha x < 1$

$$\Rightarrow f(x,0) = x^2 n(x,0) = \frac{x}{\alpha\left(1+\frac{1}{2}\alpha x\right)}$$

$$\Rightarrow \frac{x}{\alpha} = f\left(1+\frac{\alpha x}{2}\right) \Rightarrow \frac{x}{\alpha} - \frac{\alpha f x}{2} = f$$

$$x\left(1-\frac{\alpha^2 f}{2}\right) = \alpha f$$

$$x = \frac{\alpha f}{1-\frac{\alpha^2 f}{2}}$$

$$\Rightarrow \phi(f) = \frac{\alpha f}{1-\frac{\alpha^2 f}{2}}$$

$$\Rightarrow x + 2tf = \phi = \frac{\alpha f}{1-\frac{\alpha^2 f}{2}}$$

$$x\left(1-\frac{\alpha^2 f}{2}\right) + 2tf\left(1-\frac{\alpha^2 f}{2}\right) = \alpha f$$

$$x - \frac{\alpha^2 x}{2} f + 2tf - \alpha^2 t f^2 = \alpha f$$

$$\alpha^2 t f^2 + \left(\alpha + \frac{\alpha^2 x}{2} - 2t\right) f - x = 0$$

$$f = \frac{2t - \alpha - \frac{\alpha^2 x}{2} \pm \sqrt{\left(2t-\alpha-\frac{\alpha^2 x}{2}\right)^2 + 4\alpha^2 t x}}{2\alpha^2 t}$$

Need to Choose $\oplus$ sign since $f \geq 0$.

~~for a given x at t=0 we have $f_c = \alpha + \frac{\alpha^2 x}{2}$~~

BEC when $x=0$:

here $f(0,t_c) = f = \dfrac{2t_c - \alpha + \sqrt{(2t_c-\alpha)^2}}{2t_c\alpha^2} = \dfrac{2t-\alpha}{2t\alpha^2}$

$$t_c\alpha^2 f = 2t_c - \alpha \Rightarrow \alpha = (2-\alpha^2 f)t_c$$

$$t_c = \frac{\alpha}{2(1-\alpha^2 f/2)}$$

Find $f(x)$ at $t=0$:

$$f(x,t\to 0) = \frac{2 + \frac{2a(ax-2)}{a(2+ax)}}{2a^2} = \frac{2}{2a^2}\left(1 + \frac{ax-2}{2+ax}\right) =$$

$$= \frac{2ax}{a^2(2+ax)} = \frac{x}{a(1+\frac{ax}{2})} \cancel{= \frac{x}{a^2(x+\frac{2}{a})}}$$

which of course is the result we began with in the previous page.

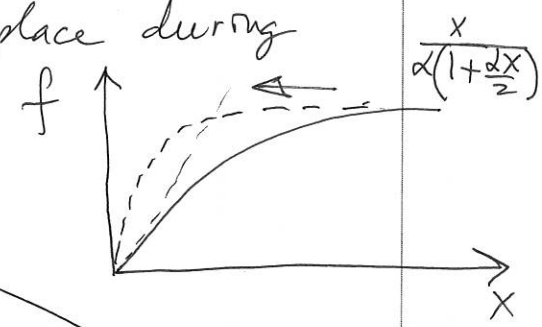
$$\to \begin{cases} \frac{x}{2a} & \text{for } x\to 0 \\ \frac{1}{a^2} & \text{for } x\to\infty \end{cases}$$

$$\Rightarrow t_c \to \frac{\alpha}{2\left(1-\frac{\alpha^2}{2}\cdot\frac{x}{2\alpha}\right)} = \frac{\alpha}{2\left(1-\frac{\alpha x}{4}\right)} \left(\text{for } x\to 0\right) = \frac{\alpha}{2}.$$

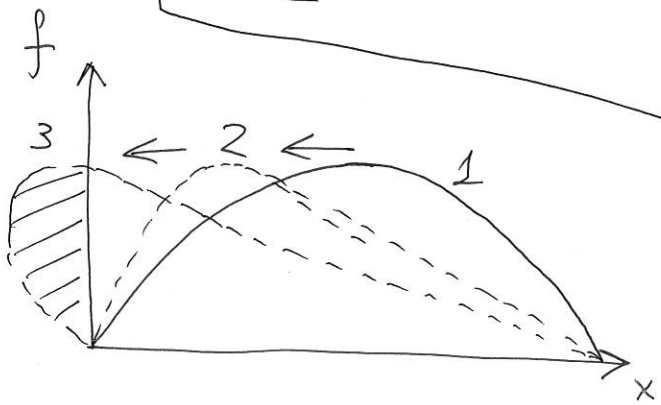
$$t_c \to \frac{\alpha}{2\left(1-\frac{\alpha^2}{2}\cdot\frac{1}{\alpha^2}\right)} = \frac{\alpha}{2\left(1-\frac{1}{2}\right)} = \alpha \quad \text{for } x\to\infty$$

In this case BEC will take place during

$$\frac{\alpha}{2} \le t \le \alpha.$$



Generally



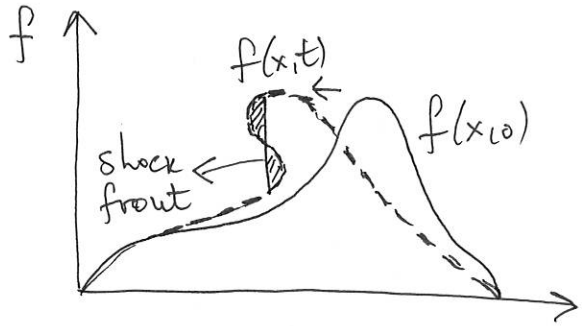
Initial distribution $f(x,0)$ [line 1]

$$x = \phi[f] - 2ft$$

means that every point of the initial curve move along (characteristic lines) straight lines parallel to x-axis, and in the <u>negative</u> x-direction. Velocity is determined by $2f$! This is denoted by <u>line 2</u>.

Since $x = \frac{\hbar\omega}{k_B T_e}$ can not be negative, therefore particles reaching the f-axis accumulate in the state $\underline{x=0}$
$\Rightarrow$ undergo Bose condensation! (This is <u>line 3</u>, shaded region)

Yet another cool feature: suppose $f(x,0)$ has an <u>inflection</u> point, i.e. $f''(x_{infl}, 0) = 0$ at some x_{infl}.



Since the velocity of approach to f-axis increases as f increases, the upper parts of the graph of $f(x,t)$ will advance toward f-axis more than the lower parts $\rightarrow$ as a result $f(x,t)$ may bend over so much that $f(x,t)$ will stop being single-valued.

This looks very similar to the <u>shock</u> wave in the 1-dim. flow of nonviscous fluid.
As the shock front is approached, $\frac{\partial n}{\partial x} \rightarrow \infty$ diverges!
And our initial approximation $n^2 \gg \frac{\partial n}{\partial x}$ breaks down.
This is analogues to the appearance of viscocity (term with $\frac{\partial n}{\partial x}$) in the photon fluid.